



Effect on first-service outcomes of a modified Double-Ovsynch protocol designed to increase fertility through insemination of cows in estrus and a targeted intervention for nonestrus cows

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ABSTRACT

The objective of this randomized controlled experiment was to evaluate the effect on first-service reproductive outcomes for lactating Holstein cows of a targeted reproductive management program including insemination in estrus (AIE) after a modified Double-Ovsynch protocol and re-enrollment of nonestrus cows in a synchronization of ovulation protocol. Lactating dairy cows ($n = 2,101$) from a commercial dairy farm fitted with neck-attached sensors for automated detection of estrus were synchronized with a Double-Ovsynch protocol up to the first PGF_{2α} (PGF-L) of the Breeding-Ovsynch portion of the protocol (GnRH, 7 d later PGF_{2α}, 3 d later GnRH, 7 d later GnRH, 7 d later PGF_{2α}, 1 d later PGF_{2α}). At PGF-L, cows were blocked by parity and type of semen used for first service and then randomly assigned to the DO-G80 ($n = 1,048$) and the DO-ED-P4Ov ($n = 1,053$) treatments. Cows in the DO-G80 treatment received GnRH for induction of ovulation at 80 h after induction of luteolysis (GnRH 80 h after PGF-L, 16 h later timed artificial insemination [TAI]). All cows with automated estrus alerts (AEA) after PGF-L and before the GnRH treatment received AIE. Cows in the DO-ED-P4Ov treatment were eligible to receive AIE for 7 d after PGF-L and, if not AIE, were submitted to a modified Ovsynch protocol with progesterone (P4) supplementation (P4-Ov: GnRH + P4 device in, 7 d later PGF_{2α} and P4 device removal, 1 d later PGF_{2α}, 32 h later GnRH, 16 h later TAI). Data were analyzed by logistic regression and ANOVA. The DO-ED-P4Ov treatment increased the proportion of cows AIE (+41%; DO-G80 46.7% and DO-ED-P4Ov 88.1%) and first-service pregnancy per artificial insemination (P/AI; +8%) compared with the DO-G80 treatment through enhanced P/AI of primiparous cows (primiparous DO-G80 50.5% and DO-ED-P4Ov 63.6%; multiparous DO-

G80 45.3% and DO-ED-P4Ov 48.5%) and P/AI in the range of 45% for nonestrus cows targeted with the P4-Ov protocol. Despite increased costs of individual artificial insemination (AI) services, the DO-ED-P4Ov treatment also had reduced costs per pregnancy. In conclusion, a targeted reproductive management program designed to increase fertility through insemination of cows in estrus after a modified Double-Ovsynch protocol and a targeted intervention for nonestrus cows might be an alternative to increase first-service fertility and reduce costs per pregnancy primarily through better performance of primiparous cows. This program could also have other herd management benefits including identifying cows of different fertility potential and the distribution of AI services over more days of the week compared with TAI. **Key words:** automated detection of estrus, synchronization of ovulation, targeted reproductive management, dairy cow

INTRODUCTION

Most of the commonly used protocols for synchronization of ovulation for timed artificial insemination (TAI) include a 56-h interval between induction of luteolysis and induction of ovulation before artificial insemination (AI; Bello et al., 2006; Souza et al., 2008; Wiltbank and Pursley, 2014). This interval increases pregnancy per AI (P/AI) after TAI compared with shorter or longer intervals (Brusveen et al., 2008) by optimizing timing of ovulation in relationship to luteolysis and pre-ovulatory follicle development. In most cows, this interval from induction of luteolysis to GnRH also prevents spontaneous ovulation and expression of estrus before insemination (Lauber and Fricke, 2024; Laplacette et al., 2025) as the GnRH-induced LH surge causes a shift from a follicular to a luteal hormone secretion pattern (Fortune et al., 2009). Thus, to benefit from the putative positive effects of expression of estrus before AI on fertility (Larimore et al., 2015; Madsen et al., 2015; Pereira et al., 2016), delaying the last GnRH of the Breeding-Ovsynch portion

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

of the Double-Ovsynch (**DO**; Pre-Ovsynch: GnRH, 7 d later PGF_{2α}, 3 d later GnRH, 7 d later Breeding-Ovsynch: GnRH, 7 d later PGF_{2α}, 1 d later PGF_{2α}, 32 h later PGF_{2α}, 16 h later GnRH) protocol was recently explored as a strategy to allow more cows express estrus before insemination (Laplacette et al., 2025).

In a randomized controlled experiment with lactating Holstein cows, Laplacette et al. (2025) demonstrated that delaying induction of ovulation from 56 to 80 h after induction of luteolysis in the DO protocol increased the proportion of cows detected in estrus through automated estrus alerts (**AEA**) before AI from ~28% to 71%. Cows inseminated in estrus (**AIE**) before induction of ovulation with GnRH (~32%) and cows with estrus detected after induction of ovulation with GnRH (~39%) had more P/AI than cows with no estrus detected (nonestrus cows) that received GnRH and TAI (LSM ~60% vs. 35% for estrus and nonestrus cows, respectively). The relatively low P/AI of the ~29% nonestrus cows that received TAI was sufficient to prevent an overall gain in fertility for the protocol with GnRH at 80 h versus the DO protocol (herein **DO-G80** protocol) with GnRH 56 h after induction of luteolysis.

In synchronization of ovulation programs with delayed induction of ovulation, expression of estrus enables self-separation of cows in groups with stark differences in fertility and physiological outcomes associated with insemination success. For example, Laplacette et al. (2025) reported that cows that expressed estrus, especially those with estrus before induction of ovulation, were more likely to have complete luteal regression, had a more favorable uterine environment (i.e., more cows had uterine tone and fluid, and had greater luminal fluid scores), more cows had a large follicle (defined as >16 mm), and the largest follicle size was greater than that of nonestrus cows. In addition, more cows with estrus ovulated a larger follicle, which resulted in more cows with a functional and larger corpus luteum (**CL**) and greater circulating concentrations of progesterone (**P4**) 6 d after ovulation. Collectively, these endocrine and ovarian function differences between estrus and nonestrus cows suggested that despite delaying the GnRH treatment to 80 h after triggering luteolysis, premature induction of ovulation could have contributed to the poor fertility of nonestrus cows. At least partially, other factors inherent to the design of the protocol with delayed induction of ovulation such as a longer period of follicle dominance due to delayed (i.e., 1 d) induction of ovulation and an overrepresentation of cows with lower reproductive potential (i.e., multiparous, high producing, and cows with health events before AI) as observed in Laplacette et al. (2025) could have also contributed to the fewer P/AI of nonestrus cows that received TAI.

To address the limitations of synchronization of ovulation protocols designed to promote estrus expression be-

fore AI, a 2-pronged approach could be used to optimize fertility. First, extending the interval for expression of estrus after induction of luteolysis could maximize the number of cows that express estrus and have spontaneous ovulation. Indeed, it has been reported that most cows expressed estrus (~85%) after synchronization with a protocol that generated ovarian function and endocrine environment similar to the DO protocol (Schilkowsky et al., 2021). Expression of estrus in most cows could enable the putative positive effects of estrus on fertility and prevent premature induction of ovulation in some cows. Enabling more cows to express estrus could also result in further separation of cows with different fertility potential and minimize the size of the nonestrus group. The latter group, however, must be re-enrolled in a synchronization of ovulation protocol effective for cows without a functional CL (Bisinotto et al., 2015; Wijma et al., 2018) because the fertility of cows that receive TAI several days after luteolysis would be likely low.

Therefore, we hypothesized that maximizing the proportion of cows inseminated at detected estrus at the end of the DO protocol and optimizing the fertility of nonestrus cows through enrollment in the Ovsynch protocol with P4 supplementation (**P4-Ov**) would increase first-service P/AI compared with a DO protocol with delayed induction of ovulation (i.e., 80 h after induction of luteolysis). Specifically, we hypothesized that the combination of more inseminations at detected estrus after induction of luteolysis during the Breeding-Ovsynch portion of DO, and optimized fertility for nonestrus cows, would increase overall P/AI through the positive effects of estrus on fertility and a lesser decline in fertility for nonestrus cows. Therefore, the primary objective of this randomized control experiment with lactating Holstein cows was to evaluate the effect on P/AI for cows submitted to first service using a DO protocol with AIE for up to 7 d after induction of luteolysis and enrollment of nonestrus cows in P4-Ov compared with a DO protocol delayed induction of ovulation for TAI. Secondary objectives were to evaluate the proportion of cows that expressed estrus and received AIE, DIM at first insemination, pregnancy loss, the proportion of cows pregnant by DIM, and first-service costs per insemination and per pregnancy.

MATERIALS AND METHODS

All experimental procedures conducted with animals were approved by the Animal Care and Use Committee of Cornell University (protocol no. 2021-0031).

Farm and Cow Management

This experiment was conducted at a commercial dairy farm located in Fort Collins, Colorado, from December of

2022 to January of 2024. The farm used DairyComp305 (Valley Agricultural Software) for dairy herd management. Cow numbers and milk yield (MY) data were retrieved using the ECON\ID command from DairyComp305.

During the experiment, the farm housed an average of 5,207 (range: 5,054–5,450) milking and 646 (range: 522–736) dry cows. Lactating cows were housed by parity (i.e., primiparous vs. multiparous) in 4-row freestall tunnel-ventilated barns with deep sand-bedded stalls and concrete alleys. Cows were fed a TMR 2×/d with ad libitum access to feed and water. Diets were formulated to meet or exceed the nutrient requirements of lactating dairy cows producing 42.0 kg/d of milk with 4.1% fat and 3.1% true protein, as indicated by the Cornell Net Carbohydrate and Protein System (CNCPS v. 6.5.5; Van Amburgh et al., 2015). Cows were milked 3×/d at approximately 8-h intervals in an external 90-point rotary milking parlor equipped with individual stall milk meters (DeLaval milk meter MM27BC, DeLaval) to capture individual cow milk weights. The average MY per cow for the milking herd during the study period was 41.3 kg/d (range: 37.7–52.2 kg/d).

All cows were fitted with a neck-attached electronic accelerometer-based sensor tag at dry-off (SenseHub Monitoring Neck Tag, Merck Animal Health). At enrollment and at the time of first service, sensor functionality was confirmed by evaluating raw physical activity and rumination data continuity (i.e., no gaps in the activity and rumination line in the cow card) for individual cows in the software interface. All estrus events detected based on AEA were recorded.

Experimental Design and Procedures

The experiment followed a randomized complete block design with parity (primiparous vs. multiparous) and type of semen assigned for first service (sex-sorted dairy Holstein and Jersey vs. conventional beef) as blocking factors. Cows were eligible for enrollment in the experiment if were of the Holsteins breed, were in the milking herd, had an automated monitoring sensor attached, and were not coded as “do not breed.”

Per the farm operating procedures, cows were eligible for enrollment in synchronization of ovulation protocols, and therefore in the experiment, daily through custom-built automated commands set in the herd management software. Cows were enrolled at different DIM depending on parity group and experimental treatment. Primiparous cows were enrolled at 68 DIM. Per herd management procedures, multiparous cows were enrolled in the synchronization of ovulation protocol at different DIM based on diagnosis of endometritis after a single examination with transrectal ultrasonography of the uterine content at 25 DIM. Endometritis was de-

finied as the presence of echogenic content in the uterine lumen. Cows not diagnosed with endometritis (73.2%) were enrolled in the DO protocol at 33 DIM, whereas cows diagnosed with endometritis (26.8% of all cows enrolled) received a PGF_{2α} treatment (500 µg of Cloprostenol sodium; Synchsure, Boehringer Ingelheim) at 25 and 39 DIM and were enrolled in the synchronization of ovulation protocol at 43 DIM.

All cows had their estrous cycle synchronized following the sequence of hormonal treatments of the DO protocol (Pre-Ovsynch: GnRH, 7 d later PGF_{2α}, 3 d later GnRH, 7 d later Breeding-Ovsynch: GnRH, 7 d later PGF_{2α}, 1 d later PGF_{2α}) up to the second PGF_{2α} treatment of the Breeding-Ovsynch portion of the protocol as originally described in Souza et al. (2008) and modified in Brusveen et al. (2009).

At induction of luteolysis with the first PGF_{2α} (herein **PGF-L**) treatment of the Breeding-Ovsynch portion of DO, cows (n = 2,101) were blocked by parity (primiparous: n = 591 [28.1%]; multiparous: n = 1,510 [71.9%]) and type of semen assigned for first service (sex-sorted: n = 775 [36.9%]; conventional beef: n = 1,326 [63.1%]) and then randomly allocated within block to the experimental treatments (Figure 1). Cows in the DO-G80 treatment (n = 1,048) with an automated estrus alert (AEA) within 80 h of PGF-L received AIE without GnRH. Cows with no AEA by 80 h after PGF-L received GnRH and TAI ~16 h later. Cows in the **DO-ED-P4Ov** treatment (n = 1,053) with an AEA within 7 d of the PGF-L were inseminated without GnRH; cows not inseminated by 7 d after PGF-L were enrolled in P4-Ov (GnRH and intravaginal P4-releasing device insertion, 7 d later PGF_{2α} and P4-releasing device removal, 1 d later PGF_{2α}, 32 h later GnRH, 16 h later TAI) as described in Wijma et al. (2018). For primiparous cows, this resulted in AIE from 92 to 95 DIM and TAI at 96 DIM for the DO-G80 treatment and AIE from 92 to 98 DIM and TAI at 109 DIM for the DO-ED-P4Ov treatment (Figure 1). For multiparous cows not diagnosed with endometritis (73.2%), AIE occurred from 57 to 60 DIM and TAI at 61 DIM for the DO-G80 treatment, whereas for the DO-ED-P4Ov treatment AIE occurred from 57 to 63 DIM and TAI at 74 DIM (Figure 1). For multiparous cows diagnosed with endometritis (26.8% of all cows enrolled), AIE occurred from 67 to 70 DIM and TAI at 71 DIM, whereas cows in the DO-ED-P4Ov treatment received AIE at 67 to 73 DIM and TAI 84 DIM. Cows were allocated to treatments automatically by the dairy herd management software (DairyComp305) using a custom-built command that randomly assigned cows to treatments within block using a random number generator. All personnel working with cows on farms were blind to the experimental treatments.

Cows were assigned to receive sex-sorted dairy or conventional beef semen based on genetic merit using

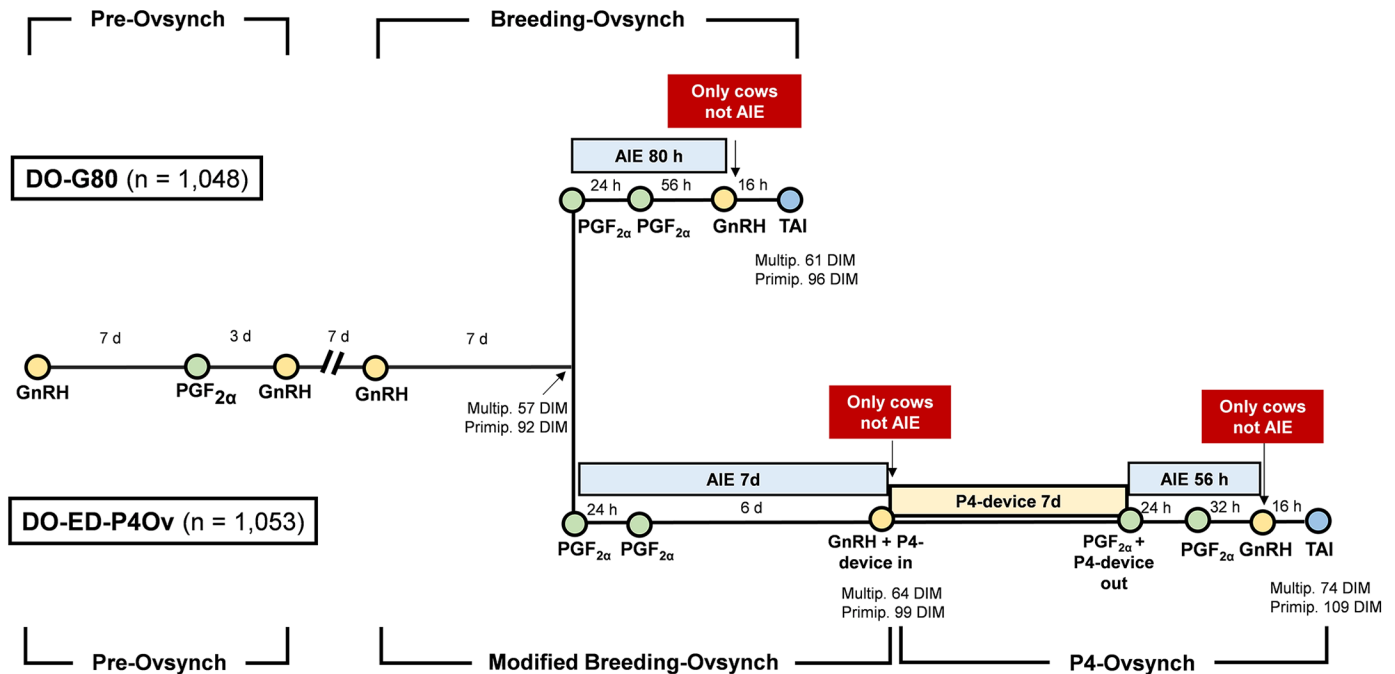


Figure 1. Graphical depiction of the experimental procedures. Cows had the estrous cycle synchronized following the sequence of hormonal treatments of a Double-Ovsynch protocol up to the second PGF_{2α} treatment of the Breeding-Ovsynch portion of the protocol (Pre-Ovsynch: GnRH, 7 d later PGF_{2α}, 3 d later GnRH, 7 d later Breeding-Ovsynch: GnRH, 7 d later PGF_{2α}, 1 d later PGF_{2α}). At induction of luteolysis with the first PGF_{2α} (PGF-L) treatment of Breeding-Ovsynch, cows blocked by parity and type of semen used were randomly assigned to the DO-G80 treatment (n = 1,048) or the DO-ED-P4Ov treatment (n = 1,053). Cows in the DO-G80 treatment with an automated estrus alert for up to 80 h after PGF-L were inseminated in estrus (AIE) without GnRH. Cows with no AEA by 80 h after PGF-L received GnRH and were timed inseminated ~16 h later. Cows in the DO-ED-P4Ov treatment with an AEA up to 7 d after PGF-L received AIE without GnRH. Cows not AIE by 7-d after PGF-L were enrolled in a modified Ovsynch protocol with P4 supplementation and 2 PGF_{2α} injections (GnRH and intravaginal P4-releasing device insertion, 7 d later PGF_{2α} and P4-releasing device removal, 1 d later PGF_{2α}, 32 h later GnRH, 16 h later TAI).

criteria defined by farm operating procedures. For the DO-G80 treatment, 28.1% (n = 294) and 71.9% (n = 754) of the cows were primiparous and multiparous, respectively, and 35.9% (n = 376) and 64.1% (n = 672) of the cows, respectively, were assigned to receive sex-sorted dairy or beef semen. For the DO-ED-P4Ov treatment, 28.2% (n = 297) and 71.8% (n = 756) of the cows were primiparous and multiparous, respectively, and 37.9% (n = 399) and 62.1% (n = 654) of the cows, respectively, were assigned to receive sex-sorted dairy or beef semen.

All GnRH treatments consisted of an i.m. injection of 100 µg of Gonadorelin diacetate (Cystorelin, Boehringer Ingelheim), PGF_{2α} treatments for synchronization of ovulation consisted of an i.m. injection of 500 µg of Cloprostenol sodium (Synchsure, Boehringer Ingelheim) and P4 supplementation was provided with an intravaginal P4-releasing device (1.38 g of P4; Eazi-Breed CIDR, Zoetis). All cows were sorted automatically after milking to receive treatments administered by farm personnel. Radio frequency ear tags were scanned using a wand reader (Workabout Pro4 Barcode scanner, Symbol) to determine the necessary treatment and log the time of each treatment. Data for treatment time was transferred to DairyComp305.

Timed AI was conducted after milking for the session closest to 16 h after the GnRH treatment before the scheduled TAI. Inseminations at detected estrus were conducted 3×/d after cows exited the milking parlor for the milking session immediately before the AEA was generated. A cow received AIE if an AEA was triggered and the heat index calculated by the DataFlow software (DataFlow II Monitoring, Merck Animal Health) was ≥36 points following internal farm standard operating procedures. The time of the onset of all AEA was recorded manually from the events tab of the DataFlow system. Before each milking session began, farm personnel in charge of reproductive management generated a list of cows with an active AEA before the beginning of the milking session and were therefore eligible for AIE. If an AEA was triggered during the milking session, the cow received AIE after the next milking session.

Insemination at detected estrus and TAI were conducted by the same 6 on-farm technicians. Pregnancy diagnosis was conducted by transrectal ultrasonography of the uterine contents and ovaries using a portable ultrasound machine with an attached linear rectal probe set at 7.5 MHz (Ibex Pro, Ibex). Pregnancy diagnosis was

conducted every day of the week by farm personnel ($n = 4$) so that the first pregnancy exam was conducted at 30 d after AI (30-d pregnancy evaluation) and confirmation of pregnancy was performed at 50 d after AI (50-d pregnancy evaluation). Pregnancy loss was recorded when a cow pregnant at 30 d after AI was not pregnant at 50 d after AI, or if AIE occurred before pregnancy confirmation at 50 d after AI.

Health Disorder Events, Season of AI, and Milk Yield Data Collection

Health disorders were diagnosed and recorded by farm personnel from calving to first service following farm operating procedures as described in Rial et al. (2024). Health events of interest for analysis were milk fever, retained placenta, metritis, clinical mastitis, displaced abomasum, indigestion, hyperketonemia (blood BHB >1.2 mmol/L at 5 DIM), clinical ketosis, and pneumonia. Concentrations of BHB in circulating blood collected by puncture of the tail vein or artery were determined using BHB strips and a strip reading device (BHBCheck Plus, PortaCheck Inc.). Cows were included in the health disorder group if had at least one health event recorded up to first service.

Season of insemination was assigned to each service based on the date of insemination. Inseminations during the warm season were those from May to September and inseminations during the cold season were those from October to April.

Accumulated MY to 100 DIM was calculated by the herd management software based on daily milk weight data from parlor milk meters (MM27BC, DeLaval). Milk yield groups (high, medium, low) were created based on tertiles of MY to 100 DIM for primiparous (medium tertile lower limit = 3,445.9 kg and upper limit = 3,840.8 kg) and multiparous cows (medium tertile lower limit = 4,703.4 kg and upper limit = 5,257.3 kg).

Reproductive Cost

The total costs of the first AI service per cow was calculated as the sum of all hormonal treatments (hormone costs and labor), insemination (semen straws and labor), and detection of estrus. The cost of GnRH and PGF_{2α} was set at \$1.53 (Cystorelin, Boehringer Ingelheim) and \$2.34 per dose (Synchsure, Boehringer Ingelheim), respectively. The cost of an intravaginal P4-releasing device was set at \$15.28 (Eazi-Breed CIDR, Zoetis). All hormone costs were calculated as the average between the higher and lower price listed in 4 domestic distributors of animal health and management related products. Labor costs for reproductive procedures were set at \$16.85/h for labor based on the national livestock worker annual average

gross wage rate reported on the farm labor report released in November 2023, by the National Agricultural Statistics Service (NASS), Agricultural Statistics Board, USDA. Labor cost for application of hormonal treatments was set at \$0.28 per injection assuming 60 injections per hour and \$0.84 per controlled internal drug release (CIDR) device application assuming the application of 20 CIDR devices per hour. The cost of a unit of semen was set at \$15 for conventional beef semen and \$35 for sex-sorted dairy breed semen. The cost to inseminate a cow was set at \$1.12 per insemination assuming 15 inseminations per hour. Estrus detection cost was set by calculating the cost per day of the sensor monitoring service per cow (\$0.09; prices based on the authors experience) multiplied by the number of days from the beginning of the synchronization of ovulation protocol (day of first GnRH treatment of DO) until the date of first service.

In addition, the total cost per pregnancy was calculated by dividing the total cost of first AI service by the P/AI within parity for each treatment.

Sample Size Calculation

The sample size required was estimated using the comparison of proportions option of the sample size calculation menu of WinPepi version 11.65 (Abramson, 2011). A total of 1,776 cows (888 per treatment) were needed to detect a significant difference in P/AI at first service of up to 6.0 percentage points in favor of the DO-ED-P4Ov treatment assuming 50% P/AI for the DO-G80 treatment (control) for a 1-tailed test with a probability of Type I error of 5% and Type II error of 20%. Approximately 15% more cows were enrolled to compensate for cows that left the herd before receiving first service.

Statistical Analysis

All statistical analyses were conducted using SAS software (version 9.4, SAS Institute Inc.), except for the sample size calculation.

Outcomes of interest represented by proportions generated from binary outcomes included the proportion of cows AIE and with estrus expression before first AI, P/AI at the 30-d and 50-d pregnancy diagnosis, and pregnancy loss. All analyses for these outcomes were conducted using logistic regression with PROC GLIMMIX fitting a binomial distribution. In addition, the total proportion of cows pregnant out of those inseminated by the latest DIM at which first service occurred for cows enrolled in the experiment (includes first and second AI services for cows re-inseminated by the end-point DIM) was compared. This analysis was conducted to evaluate if the different length of treatments affected the pregnancy dynamics by the time that all cows had the opportunity to receive first service.

Thus, cows with a first service or re-insemination by the DIM at which cows received first service in the longest treatment were included (i.e., 109 DIM for primiparous and 74 and 84 DIM for multiparous cows without and with endometritis diagnosis in DO-ED-P4Ov, respectively). Cows were considered pregnant for this analysis if confirmed pregnant at 50 d after insemination.

Continuous outcomes including time from induction of luteolysis before AI to the onset of AEA and to AIE, DIM at first service, and reproductive costs were analyzed by ANOVA using PROC MIXED. Cows with no AEA and cows that received TAI were excluded from the analysis of time to AEA and AIE because the goal of the analysis was to evaluate time to event for cows with the event only. Assumptions of normality of residuals, linear relationship, and homoscedasticity of variance for residuals in linear regression models were assessed by evaluating normal probability plots and plots of residuals versus predicted values.

All models built for analyses with logistic regression and ANOVA included treatment (DO-G80 vs. DO-ED-P4Ov) and parity (primiparous vs. multiparous) as fixed effects. For models that evaluated P/AI and pregnancy loss, the type of semen used (sexed dairy vs. conventional beef) was included as a fixed effect and AI technician as random effect. Covariates offered to models included: MY group (high, medium, low), season of AI (cool vs. warm), and a categorical variable that indicated health disorder group (0 = no disorder; 1 = one or more health disorder events). For all models, the interactions between treatment and parity, season of AI, MY group, and health disorder group were offered. In addition, models for evaluating P/AI and pregnancy loss were offered the interaction between treatment by type of semen used and parity by type of semen used. The final model for each outcome of interest was selected by manual backward elimination of covariates and their interactions if $P > 0.10$.

Explanatory variables and interactions were considered significant if $P \leq 0.05$, whereas $0.05 < P \leq 0.10$ were considered a statistical tendency. When appropriate, the LSD post hoc mean separation test was used to determine differences between LSM. No adjustments for multiple testing across outcomes was conducted. Unless otherwise stated, results are reported as LSM (95% CI) and odds ratios (OR; 95% CI) for proportions or as LSM $\pm$ SEM for continuous outcomes.

Stochastic Analysis of First Service Costs and Costs Per Pregnancy

Stochastic Monte Carlo simulation models were developed in Python (v3.12.7) using the scipy.stats module (Virtanen et al., 2020) of the SciPy library (v1.13.1) to estimate differences between treatments for first AI

service costs and costs per pregnancy at first AI under varying market scenarios for input costs. Outcomes of interest were the distribution of the differences in total costs between the DO-G80 and DO-ED-P4Ov treatments and the contribution of individual input costs to the total variation in the differences between treatments. Fixed inputs for models were the number of GnRH and PGF_{2 α} treatments administered, intravaginal P4-devices, sex-sorted dairy and conventional beef semen straws used, the total days at risk for estrus detection, and the hours of labor needed to implement each experimental treatment (Supplemental Table S3, see Notes).

Stochasticity was introduced for labor costs (\$/h), the cost of GnRH and PGF_{2 α} treatments (\$/dose), intravaginal P4 devices (\$/device), semen (\$/straw), and the cost of automated estrus detection (\$/d) in every iteration of the simulation. Values for all costs were varied by $\pm 20\%$ from the price defined for this experiment (i.e., base value; Supplemental Table S4, see Notes). Because no large dataset was available to set actual distributions for input prices, a triangular probability distribution (Back et al., 2000) centered around the base value with a variability of 20% was used to define minimum and maximum values (SciPy's triang distribution) in the simulations. Ten thousand iterations with replacement were run and recorded using the rvs() method of SciPy's triang in Python. For each iteration, the difference in the total costs of first AI service and cost per pregnancy between treatments (output) was calculated by subtracting the cost for DO-ED-P4Ov and the cost of the DO-G80 treatment.

To evaluate the contribution of each input to the uncertainty of the differences between treatments, a sensitivity analysis was conducted using a structured Monte Carlo sampling method (Saltelli sequences) to compute Sobol indices (Saltelli et al., 2004; Hart et al., 2017) with the saltelli sample function and the Sobol analyze function of the Salib library (v1.5.2, Iwanaga et al., 2022). First-order (S1) indices evaluated the contribution of every input in isolation, whereas total effect indices accounted for the effect of the input including interactions with other inputs.

RESULTS

A diagram describing the flow of study units across the experiment is included in Figure 2. Out of 2,190 cows available for enrollment, 2,101 were eligible because were Holstein cows that calved in the farm where the experiment was conducted. Sixty-three Jersey or cross-bred cows and 26 cows that calved at another farm were excluded. Out of the 2,101 cows enrolled, 2,062 cows (98.1% of the cows enrolled) received first service per the experimental procedures and were included in analyses of insemination dynamics and expression of estrus before AI. Reasons for exclusion were failure to follow

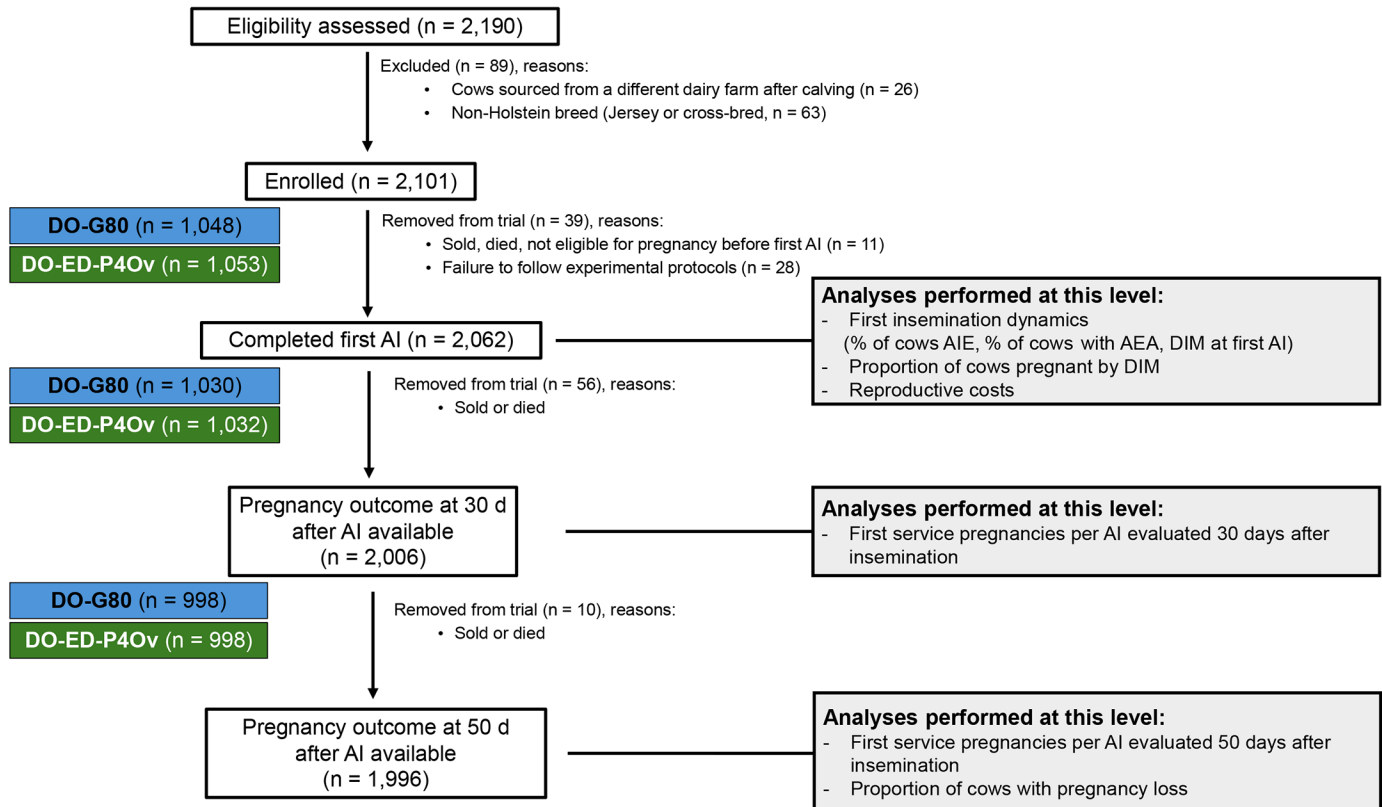


Figure 2. Flow diagram of study units at each level of the organization structure with a description of the reasons for exclusion or withdrawal from the trial and analyses performed at each stage of the experiment. Lactating Holstein cows were randomized to receive first insemination with a Double-Ovsynch protocol with an 80-h interval between induction of luteolysis and induction of ovulation (DO-G80) or a targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrus cows to an Ovsynch protocol with P4 supplementation (DO-ED-P4Ov). AIE = artificial insemination at detected estrus; AEA = automated estrus alert; TAI = timed artificial insemination.

experimental protocols (n = 28/39; i.e., errors during synchronization, wrong DIM at enrollment for synchronization), sale, death, or coded as “do not breed” (n = 11/39). Also, 2,006 (95.5% of the cows enrolled) and 1,996 (95.0% of the cows enrolled) cows had a first-service outcome at the 30-d and 50-d pregnancy evaluation, respectively. All cows that received first service were included in analyses of insemination dynamics, proportion of cows pregnant by specific DIM, and costs of insemination per first service and per pregnancy. All cows that received first service and had the outcome for the pregnancy evaluation at 30 d were included in the analysis of P/AI at first service at 30 d, and all cows that received first service and had the pregnancy outcome at 30 and 50 d were included in analyses of P/AI at reconfirmation of pregnancy. These cows were also included in the analysis of the proportion of cows with pregnancy loss. Data for the distribution of cows by parity, type of semen assigned for AI, fertile for MY, health disorder events, and season of AI for first services are included in Supplemental Table S1 (see Notes).

Insemination and Expression of Estrus Dynamics

Cows Inseminated at Detected Estrus. There was an interaction between treatment and MY group for the proportion of cows AIE ($P = 0.02$; Figure 3A) because for the DO-G80 treatment, more cows received AIE in the low (53.3% [47.8%, 58.8%], reference) than the high MY group (40.1% [34.8%, 45.7%], OR = 0.6 [0.4, 0.8]). The proportion of cows AIE for the medium MY group (46.7% [41.2%, 52.3%], OR = 0.8 [0.6, 1.4]) in the DO-G80 treatment did not differ from the other 2 groups. No differences in the proportion of cows AIE were observed among MY groups for cows in the DO-ED-P4Ov treatment (OR low MY = 6.1 [4.1, 8.9], OR medium MY = 5.7 [3.9, 8.4], and OR high MY = 7.8 [5.1, 11.9]). Overall, more cows received AIE in the DO-ED-P4Ov than the DO-G80 treatment (OR = 8.5 [6.7, 10.7], Figure 3B), and more primiparous (74.9% [70.8%, 78.7%]) than multiparous cows received AIE (68.4% [65.5%, 71.2%], OR = 0.7 [0.6, 0.9], $P = 0.007$). There was also a tendency for an interaction between

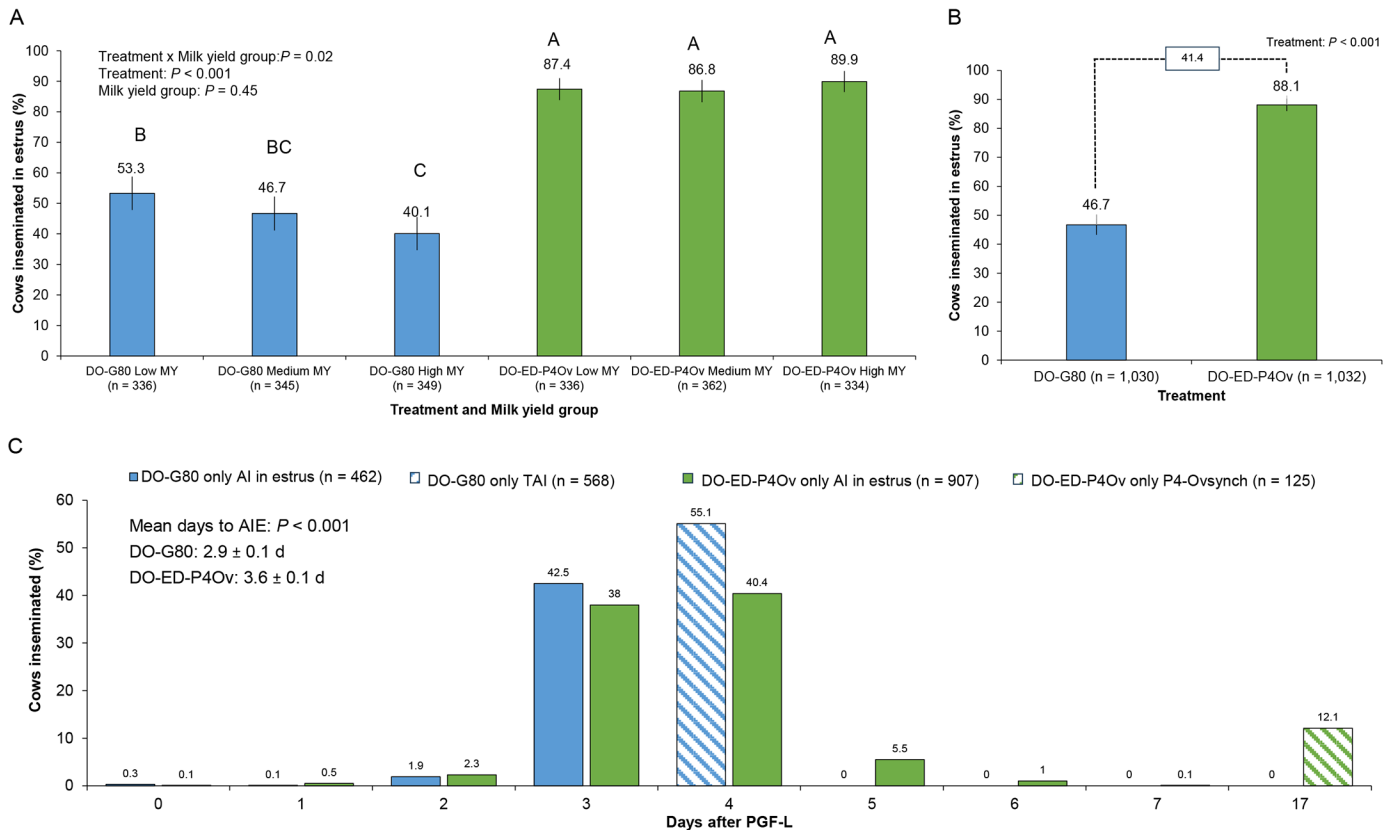


Figure 3. Percentage of cows inseminated in estrus (AIE) based on milk yield (MY) groups (A) and overall (B) for cows synchronized with the Double-Ovsynch protocol with an 80-h interval between induction of luteolysis and induction of ovulation (DO-G80, blue bars) and a targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrus cows to an Ovsynch protocol with P4 supplementation (DO-ED-P4Ov, green bars). Data are presented as LSM and 95% CI. Capital letters in panel A indicate that means for levels of a variable with different superscripts differ ($P \leq 0.05$) based on the LSD post hoc mean separation test. (C) Percentage of cows out of the total in each treatment (DO-G80: $n = 1,030$ and DO-ED-P4Ov: $n = 1,032$) that received AIE (solid bars) by day after the PGF $_{2\alpha}$ of the Breeding-Ovsynch protocol (PGF-L) and percentage of cows that received timed AI (TAI, bars with pattern) for the DO-G80 (received GnRH at 80 h and TAI because no estrus was detected) and the DO-ED-P4Ov treatment. Cows in the DO-ED-P4Ov treatment with no estrus during the 7-d period after PGF-L were submitted to an Ovsynch protocol with P4 supplementation. Data are presented as arithmetic means. For simplification, cows AIE during P4-Ov (11/1,032, 1.1%) are included in the bar corresponding to TAI for the DO-ED-P4Ov treatment.

treatment and the occurrence of health disorders before AI ($P = 0.06$) for the proportion of cows AIE.

Cows received AIE earlier in the DO-G80 (2.9 ± 0.1 d, median = 3, range = 0–3) than in the DO-ED-P4Ov (3.6 ± 0.1 d, median = 4, range = 0–7, $P < 0.001$) treatment but there was no effect of parity ($P = 0.34$) or the treatment by parity interaction ($P > 0.10$) for timing of AIE. The distribution of inseminations at detected estrus per day after the PGF-L treatment for all cows is reported in Figure 3C and by parity group in Supplemental Figure S1 (see Notes).

For the DO-ED-P4Ov treatment, 12.1% of the cows ($n = 125/1,032$) were submitted to the P4-Ov protocol, and 8.8% of these cows received AIE before the second GnRH treatment.

DIM at First Service. There was an interaction ($P = 0.02$) between treatment and health disorder group

for DIM at first service, because cows in the DO-G80 treatment without health disorder events were inseminated earlier (78.9 ± 0.2 DIM) than cows in the DO-G80 treatment with health disorder events (80.5 ± 0.3 DIM) and cows in the DO-ED-P4Ov treatment without health events (80.3 ± 0.2 DIM). Cows with health disorders in the DO-ED-P4Ov treatment were inseminated later (82.9 ± 0.3 DIM, $P = 0.02$) than the other 3 groups. Overall, cows in the DO-G80 (79.7 ± 0.2 DIM) treatment were inseminated earlier than cows in the DO-ED-P4Ov (81.6 ± 0.2 DIM) treatment. Cows were inseminated earlier in the cold (80.3 ± 0.1 DIM) than the warm season (81.0 ± 0.2 DIM, $P < 0.001$), and by design primiparous cows (96.7 ± 0.2 DIM) were inseminated later than multiparous cows (64.6 ± 0.1 DIM, $P < 0.001$).

Cows with AEA Before AI. More cows in the DO-ED-P4Ov (OR = 2.1 [1.7, 2.7]) than in the DO-G80

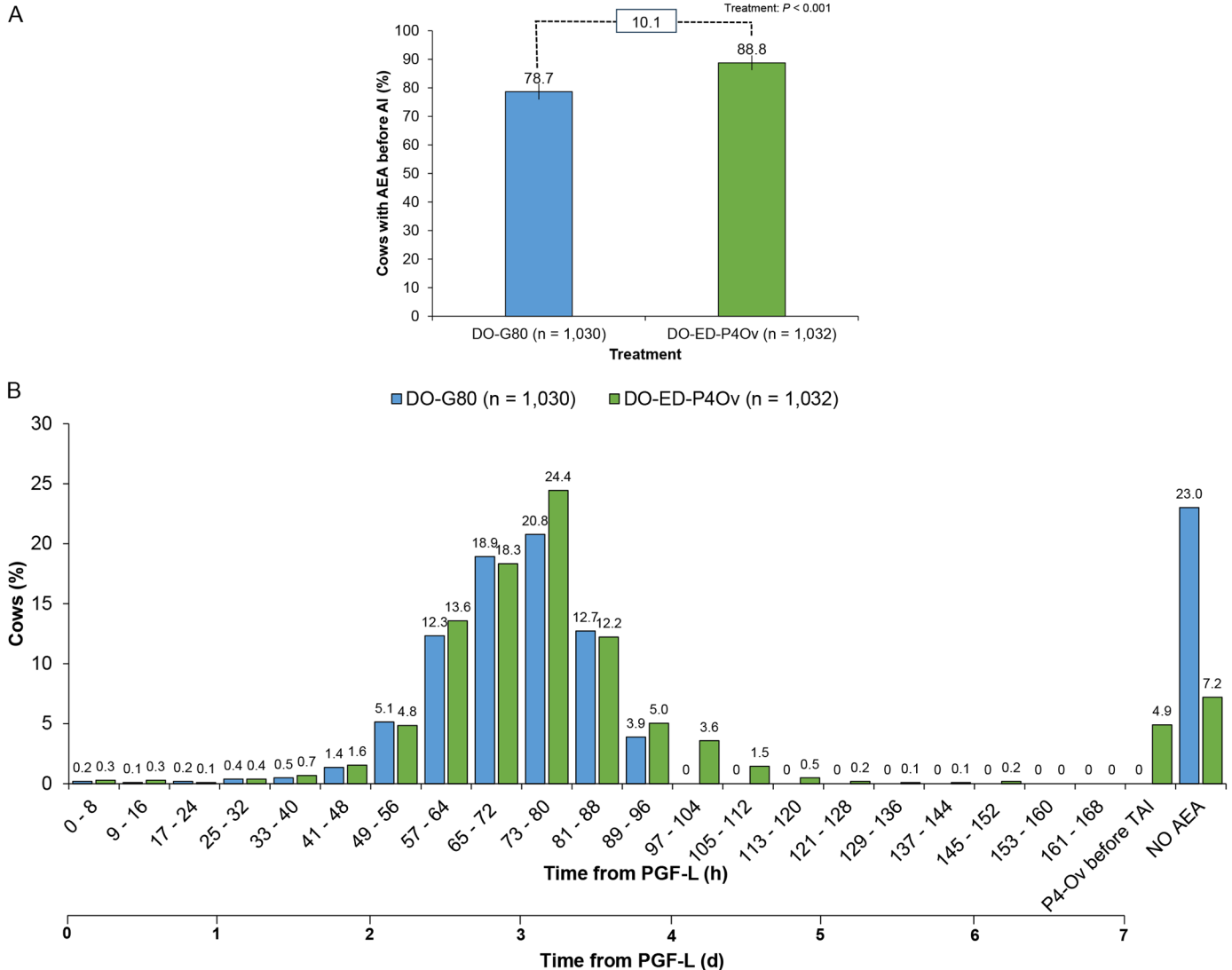


Figure 4. (A) Percentage of cows with automated estrus alerts before insemination for cows synchronized with the Double-Ovsynch protocol with 80 h between induction of luteolysis and induction of ovulation (DO-G80, blue bars) and a targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrous cows to an Ovsynch protocol with P4 supplementation (DO-ED-P4Ov, green bars). Data are presented as LSM and 95% CI. (B) Distribution of the time of the onset of AEA between induction of luteolysis (PGF-L: first PGF_{2α} of the Breeding-Ovsynch protocol) and the time of insemination for lactating dairy cows enrolled DO-G80 (blue bars) or the DO-ED-P4Ov (green bars) treatment.

treatment (reference, $P < 0.001$) expressed estrus as determined by AEA before AI (Figure 4A). In addition, more primiparous (87.6% [84.6%, 90.1%], reference) than multiparous cows (80.6% [78.3%, 82.6%], OR = 0.6 [0.4, 0.8], $P < 0.001$) and more cows without (86.2% [84.1%, 88.0%], reference) than with health disorders before first AI (82.5% [79.3%, 85.4%], OR = 0.8 [0.6, 0.9], $P = 0.02$) had estrus before AI.

In the G80 treatment, 44.9% (462/1,030) of the cows had AEA before AI and received AIE, 32.2% (332/1,030) had AEA before AI and received TAI, and 22.9% (236/1,030) did not have AEA and received TAI.

Out the 12.1% of cows in the DO-ED-P4Ov treatment that were enrolled in the P4-Ov protocol, 40.8% (51/125) had AEA before AI at the end of P4-Ov. Thus, 92.8% of all cows in the DO-ED-P4Ov treatment had an AEA before first service (i.e., 87.9% within 7 d of PGF-L and 4.9% during P4-Ov).

The frequency distribution of AEA after induction of luteolysis for 1,695 cows with data for the onset of the AEA alert after the PGF-L treatment is presented in Figure 4B. Cows in the DO-ED-P4Ov treatment submitted to the P4-Ov protocol that expressed estrus before TAI were not included in the analysis of timing of the onset

Table 1. Pregnancies per AI at the first pregnancy evaluation (30 d; primary outcome of interest for the experiment) and reconfirmation of pregnancy (50 d) for 2 treatments; results are presented as LSM [95% CI], unadjusted means (n/n), odds ratios for cows grouped by treatment and parity (primiparous vs. multiparous cows)

Item, %	Treatment ¹				P-value		
	DO-G80		DO-ED-P4Ov		Trt	Parity	Trt × Parity
	Primiparous	Multiparous	Primiparous	Multiparous			
P/AI 30-d pregnancy evaluation	50.5 ^b [43.9, 57.2] 52.8 (151/286) 1.0	45.3 ^b [40.3, 50.3] 46.8 (335/719) 0.8 [0.6, 0.9]	63.6 ^a [56.8, 69.9] 66.8 (189/283) 1.7 [1.3, 2.5]	48.5 ^b [43.5, 53.5] 51.0 (368/721) 0.9 [0.7, 1.2]	0.001 ²	<0.001	0.04
P/AI 50-d pregnancy evaluation	47.9 [41.0, 54.8] 50.4 (144/286) 1.0	41.0 [35.8, 46.4] 41.2 (293/712) 0.8 [0.6, 1.0]	59.2 [52.0, 66.1] 62.1 (175/282) 1.7 [1.1, 2.5]	42.4 [37.2, 47.8] 45.5 (326/716) 0.8 [0.6, 1.1]	0.01 ³	<0.001	0.06

^{a,b}Means for levels of a variable with different superscripts differ ($P \leq 0.05$) based on the LSD post hoc mean separation test.

¹DO-G80 = cows synchronized with the Double-Ovsynch protocol with an 80-h interval between induction of luteolysis and induction of ovulation. DO-ED-P4Ov = targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrous cows to an Ovsynch protocol with P4 supplementation.

²P/AI at the pregnancy evaluation at 30 d were 56.2%, [51.3%, 61.0%] and 47.9% [43.2%, 52.6%] for the DO-ED-P4Ov and the DO-G80 treatment, respectively (OR DO-ED-P4Ov = 1.4 [1.1, 1.7], referent: DO-G80).

³P/AI at the pregnancy evaluation at 50 d were 50.8% [45.6%, 56.1%] and 44.4% [39.4%, 49.5%] for the DO-ED-P4Ov and the DO-G80 treatment, respectively. (OR DO-ED-P4Ov = 1.3 [1.1, 1.7], referent: DO-G80).

of AEA. There was a treatment by parity interaction ($P = 0.03$) for the onset of the AEA relative to the PGF-L treatment, such that there was no difference between treatments for primiparous cows (DO-G80 = 68.5 ± 0.9 h and DO-ED-P4Ov = 69.0 ± 0.9 h), whereas the onset of the AEA was earlier for multiparous cows in the DO-G80 (72.0 ± 0.6 h) than the DO-ED-P4Ov (75.7 ± 0.5 h) treatment. Overall, cows in the DO-G80 treatment (70.3 ± 0.5 h) had the onset of AEA earlier ($P = 0.005$) than cows in the DO-ED-P4Ov (72.4 ± 0.5 h) treatment. The onset of AEA was earlier for cows in the low (69.7 ± 0.6 h) than the medium (71.8 ± 0.6 h) and high MY group (72.4 ± 0.6 h, $P = 0.003$).

P/AI and Pregnancy Loss. There was a treatment by parity interaction for P/AI at 30 d after AI (i.e., primary outcome of interest for the experiment) because primiparous cows in the DO-ED-P4Ov treatment had more P/AI than primiparous cows in the DO-G80 treatment, whereas P/AI did not differ by treatment for multiparous cows ($P = 0.04$, Table 1). Overall, the DO-ED-P4Ov (56.2%, [51.3%, 61.0%], OR = 1.4 [1.1, 1.7]) had more P/AI ($P = 0.001$) than the DO-G80 treatment (47.9% [43.2%, 52.6%]). Moreover, cows without than with health disorders ($P < 0.001$, Supplemental Table S1), and cows in the low than in the high MY group had more P/AI ($P = 0.009$; the medium MY group did not differ from high or low, Supplemental Table S2, see Notes). There was no effect of semen type on P/AI ($P = 0.18$).

For cows in the DO-G80 treatment, more P/AI were observed for those that received AIE (53.8% [48.8%,

58.8%], OR = 2.6 [1.8, 3.6]) and for cows that received TAI and had AEA (49.6% [43.8%, 55.4%], OR = 2.2 [1.5, 3.1]) than for cows that received TAI and had no AEA (31.1% [25.0%, 37.8%], OR reference, $P < 0.001$). For the DO-ED-P4Ov treatment it was not possible to compare P/AI for cows that received TAI and had or did not have estrus before AI because of the few cows submitted to P4-Ov. For descriptive purposes, P/AI were 56.9% (503/884) and 45.0% (54/120) for cows that received AIE within 7 d of the PGF-L treatment and cows inseminated after enrollment in the P4-Ov protocol. For the cows that received the P4-Ov protocol and received AIE, P/AI were 80% (8/10). Note that one cow inseminated in estrus did not have pregnancy outcome because it left the herd before the outcome could be recorded. Pregnancies per AI per day after the PGF-L treatment for cows that received AIE in the DO-G80 and the DO-ED-P4Ov treatment are presented in Table 2.

At 50 d after AI, there was a tendency for an interaction between treatment and parity ($P = 0.06$, Table 1) and an overall effect of treatment ($P = 0.01$, OR = 1.3 [1.1, 1.7]) consistent with the effect observed at the 30-d pregnancy evaluation. All the same effects of covariates on P/AI observed at 30 d after AI were observed for P/AI evaluated at 50 d after AI (Supplemental Table S2). In addition, there was a tendency for an interaction between the treatment and type of semen used ($P = 0.06$) and between parity and semen type ($P = 0.08$, Supplemental Table S2).

There was an interaction between treatment and the type of semen used on pregnancy loss ($P = 0.03$) be-

Table 2. Absolute and cumulative pregnancies per AI (P/AI) at pregnancy evaluation at 30 d for cows in the DO-G80 and DO-ED-P4Ov treatments; data are presented as arithmetic means (n/n) for all cows and per parity groups (primiparous and multiparous cows)¹

Item	AIE ≤2 d after PGF-L	AIE 3 d after PGF-L	AIE 4 d after PGF-L	AIE ≥5 d after PGF-L	TAI
First AI P/AI, % (n/n)					
DO-G80 overall	50.0 (12/24)	55.2 (237/429)	—	—	43.2 (237/549)
DO-ED-P4Ov overall	57.1 (16/28)	58.3 (224/384)	59.4 (241/406)	33.3 (22/66)	45.0 (54/120)
DO-G80 primiparous	42.9 (3/7)	60.0 (81/135)	—	—	46.5 (67/144)
DO-ED-P4Ov primiparous	42.9 (3/7)	65.2 (88/135)	75.5 (80/106)	50.0 (5/10)	52.0 (13/25)
DO-G80 multiparous	52.9 (9/17)	53.1 (156/294)	—	—	42.0 (170/405)
DO-ED-P4Ov multiparous	61.9 (13/21)	54.6 (136/249)	53.7 (161/300)	30.4 (17/56)	43.0 (41/95)
Cumulative P/AI, % (n/n)					
DO-G80 overall	50.0 (12/24)	55.0 (249/453)	—	—	48.5 (486/1,002)
DO-ED-P4Ov overall	57.1 (16/28)	58.2 (240/412)	58.8 (481/818)	56.9 (503/884)	55.5 (557/1,004)
DO-G80 primiparous	42.9 (3/7)	59.2 (84/142)	—	—	52.8 (151/286)
DO-ED-P4Ov primiparous	42.9 (3/7)	64.1 (91/142)	68.9 (171/248)	68.2 (176/258)	66.8 (189/283)
DO-G80 multiparous	52.9 (9/17)	53.0 (165/311)	—	—	46.8 (335/716)
DO-ED-P4Ov multiparous	61.9 (13/21)	55.2 (149/270)	54.4 (310/570)	52.2 (327/626)	51.0 (368/721)

¹Based on the day of AI after the first PGF_{2α} of the Breeding-Ovsynch portion of the Double-Ovsynch protocol (PGF-L) for cows inseminated in estrus (AIE) and for cows that received TAI for the DO-G80 (Double-Ovsynch protocol with 80 h between induction of luteolysis and induction of ovulation) and the DO-ED-P4Ov treatment (cows received AIE for 7 d after induction of luteolysis in the Double-Ovsynch protocol and nonestrus cows were submitted to an Ovsynch protocol with P4 supplementation). For simplification, cows AIE during P4-Ov (1.1%) were included in the TAI group for the DO-ED-P4Ov treatment.

cause cows in the DO-G80 treatment inseminated with sex-sorted semen (5.9% [3.1%, 11.0%], arithmetic mean [n/n] = 5.7 [10/177], OR reference) tended to have fewer pregnancy losses than cows in the DO-ED-P4Ov treatment inseminated with sex-sorted semen (11.5% [7.3%, 17.7%], arithmetic mean [n/n] = 10.3 [21/203], OR = 2.0 [0.9, 5.0]). In contrast, pregnancy losses for cows inseminated with beef semen did not differ between treatments (DO-G80 = 10.1% [6.8%, 14.6%], arithmetic mean [n/n] = 11.5 [35/305], OR = 1.7 [0.8, 3.3] and DO-ED-P4Ov = 7.4% [4.8%, 11.0%], arithmetic mean [n/n] = 8.3 [29/348], OR = 1.3 [0.6, 2.0]). These observations should be taken with caution because of the small sample size. Overall, no effect ($P = 0.44$) of treatment on pregnancy losses for first AI was observed (DO-G80 = 7.8% [5.2%, 11.4%], OR reference and DO-ED-P4Ov = 9.2% [6.6%, 12.7%], OR = 1.3 [0.8, 2.0]). There was, however, an effect of parity ($P = 0.02$) and an effect of health disorder before first AI ($P = 0.05$) whereby fewer primiparous than multiparous cows and fewer cows without than with health disorders had pregnancy losses (Supplemental Table S2).

Proportion of Cows Pregnant by DIM. The proportion of cows pregnant by the time that all cows received first service (n = 2,062; includes first and second AI services) was greater ($P = 0.01$) for the DO-ED-P4Ov (49.3% [45.8%, 52.7%], OR = 1.3 [1.1, 1.4]) than the DO-G80 treatment (43.8% [40.5%, 47.2%]). For the DO-G80 treatment 19 cows were re-inseminated and 4 became pregnant (21% P/AI), whereas for the DO-ED-P4Ov treatment 7 cows were re-inseminated and 2 became pregnant (29% P/AI) by the end of the observation period. Sixty-six cows that received first service and did not have a pregnancy test outcome 50 d after

AI due to sale before the 30-d (DO-G80: n = 28; DO-ED-P4Ov: n = 28) or the 50-d exam (DO-G80: n = 4; DO-ED-P4Ov: n = 6) were included in this analysis but were assumed to be nonpregnant. More primiparous ($P < 0.001$) than multiparous cows, and more cows without ($P < 0.001$) than with health disorders before first AI were pregnant by the end of the first service period. Moreover, more cows in the low than in the high MY group were ($P = 0.03$) pregnant by the end of the period, whereas cows in the medium MY group did not differ from the other groups. Effects of covariates on the percentage of cows pregnant by the end of first service are reported in Supplemental Table S2.

Costs of First Service and Costs per Pregnancy. The mean cost of first service was greater ($P < 0.001$, Table 3) for the DO-ED-P4Ov (\$42.60 ± 0.4) than the DO-G80 (\$40.20 ± 0.4) treatment. Reproductive costs were greater for primiparous (\$43.70 ± 0.4) than multiparous cows (\$39.50 ± 0.3, $P < 0.001$), and for cows in the high (\$45.00 ± 0.4, $P < 0.001$) than in the medium (\$42.00 ± 0.4) and low (\$37.20 ± 0.4) MY groups. Cows in the medium MY group had greater first-service costs than cows in the low MY group. In addition, first-service costs were greater for cows inseminated during the cold (\$43.10 ± 0.3, $P < 0.001$) than the warm season (\$39.70 ± 0.4), and for cows without (\$42.60 ± 0.3, $P < 0.001$) than with health disorders (\$40.20 ± 0.4).

When analyzed separately, the costs of synchronization of ovulation ($P < 0.001$) and detection of estrus ($P < 0.001$) were greater for the DO-ED-P4Ov than for the DO-G80 treatment, whereas insemination costs did not differ ($P = 0.57$). There was an interaction between treatment and health disorder group before first AI on synchronization of ovulation ($P = 0.01$) and detection of estrus ($P = 0.02$)

Table 3. First-service costs per cow and per pregnancy for cows in 2 treatment groups (and further separated by primiparous and multiparous cows); costs within parity are presented if the interaction between parity and treatment on reproductive costs was significant^{1,2}

Item	Treatment				P-value			
	DO-G80 (n = 1,030)		DO-ED-P4Ov (n = 1,032)		Diff. ³	Trt	Parity	Trt × Parity
	Primiparous (n = 291)	Multiparous (n = 739)	Primiparous (n = 292)	Multiparous (n = 740)				
Total cost per cow ⁴ (\$)	40.2 ± 0.4 (40.2 ± 9.7)		42.6 ± 0.4 (42.7 ± 12.5)		2.4 2.5	<0.001	<0.001	—
Total cost per pregnancy ⁵ (\$)	81.1 ^a ± 1.2 (83.5 ± 18.8)	80.8 ^a ± 0.8 (82.6 ± 19.6)	66.0 ^b ± 1.2 (67.9 ± 17.8)	79.5 ^a ± 0.8 (81.5 ± 24.5)	—	<0.001	<0.001	<0.001
	81.1 ± 0.7 (82.8 ± 19.4)		72.7 ± 0.7 (77.7 ± 23.3)		8.4 5.1			

^{a,b}Means for levels of a variable with different superscripts differ ($P \leq 0.05$) based on the LSD post hoc mean separation test.

¹DO-G80 = cows submitted to first insemination with the Double-Ovsynch protocol with 80 h between induction of luteolysis and induction of ovulation; DO-ED-P4Ov = targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrus cows to an Ovsynch protocol with P4 supplementation.

²Results are reported as LSM ± SEM and arithmetic means ($\bar{X} \pm SD$).

³Numerical difference, in dollars, for each cost item.

⁴The costs of the first AI service were calculated as the sum of the costs of synchronization, detection of estrus, and insemination for each cow. Mean for the treatments were presented because there was no interaction between treatment (Trt) and parity.

⁵The costs per pregnancy generated after the first AI service were calculated by dividing the total costs of first AI service by P/AI within parity for each treatment.

costs because cows in the DO-G80 treatment had reduced synchronization of ovulation costs regardless of health disorder occurrence before AI compared with cows in the DO-ED-P4Ov treatment (synchronization cost: health disorder = $\$14.20 \pm 0.3$ vs. no health disorder = $\$14.30 \pm 0.2$; estrus detection cost: health disorder = $\$2.48 \pm 0.02$ vs. no health disorder = $\$2.48 \pm 0.02$). In the DO-ED-P4Ov treatment, cows with health events had greater synchronization ($\$17.00 \pm 0.3$) and detection of estrus costs ($\$2.65 \pm 0.02$) than cows without health disorders ($\$15.80 \pm 0.2$ and $\$2.59 \pm 0.02$). In addition, there was a tendency ($P = 0.07$) and an effect ($P = 0.02$) for the interaction between treatment and parity group on synchronization of ovulation and detection of estrus costs, respectively. Cows in the DO-G80 treatment had reduced synchronization of ovulation (primiparous = $\$14.20 \pm 0.3$ vs. multiparous = $\$14.30 \pm 0.2$) and detection of estrus costs (primiparous = $\$2.47 \pm 0.02$ vs. multiparous = $\$2.48 \pm 0.01$) than cows in the DO-ED-P4Ov treatment regardless of parity group. Multiparous cows in the DO-ED-P4Ov treatment had greater synchronization ($\$16.9 \pm 0.3$) and estrus costs ($\2.66 ± 0.01) than primiparous cows ($\$15.7 \pm 0.3$ and $\$2.59 \pm 0.02$, respectively).

There was an interaction between treatment and parity group for the total costs per pregnancy generated after first AI ($P < 0.001$, Table 3) because primiparous cows in the DO-ED-P4Ov treatment had reduced costs compared with primiparous cows in the DO-ED-P4Ov treatment and multiparous cows in both treatments. There were no differences between treatments for multiparous cows or primiparous cows in the DO-G80 treat-

ment and multiparous cows in both treatments. Overall mean costs per pregnancy for first service were $\$8.40$ greater ($P < 0.001$, Table 3) for the DO-G80 than the DO-ED-P4Ov treatment (Table 3). Costs per pregnancy were lower for the warm ($\$73.60 \pm 0.8$) than the cold season ($\80.20 ± 0.6, $P < 0.001$) season, for cows in the low ($\$69.10 \pm 0.8$) than in the middle ($\77.80 ± 0.8) and high MY groups ($\$83.90 \pm 0.8$, $P < 0.001$), and for cows with ($\$74.60 \pm 0.8$) than without health disorders ($\$79.20 \pm 0.6$, $P < 0.001$).

Stochastic Analysis. Differences in first-service costs (DO-ED-P4Ov minus DO-G80) under diverse market prices scenarios were always in favor of the DO-G80 treatment (Figure 5A). The cost of the intravaginal P4-releasing devices was the main contributor to the total variance (Figure 5B). Conversely, differences in costs per pregnancy at first AI under diverse pricing conditions were always in favor of the DO-ED-P4Ov treatment (Figure 5C). The main contributors to the total variance were the cost of the intravaginal P4-releasing devices, followed by the cost of conventional beef and sex-sorted semen, and the cost of GnRH treatments (Figure 5D).

DISCUSSION

In support of our hypothesis, a targeted reproductive management (TRM) program aimed at promoting expression of estrus at the end of synchronization of ovulation with the DO protocol resulted in more cows inseminated in estrus and for primiparous cows increased P/AI compared with the DO-G80 protocol. This P/AI

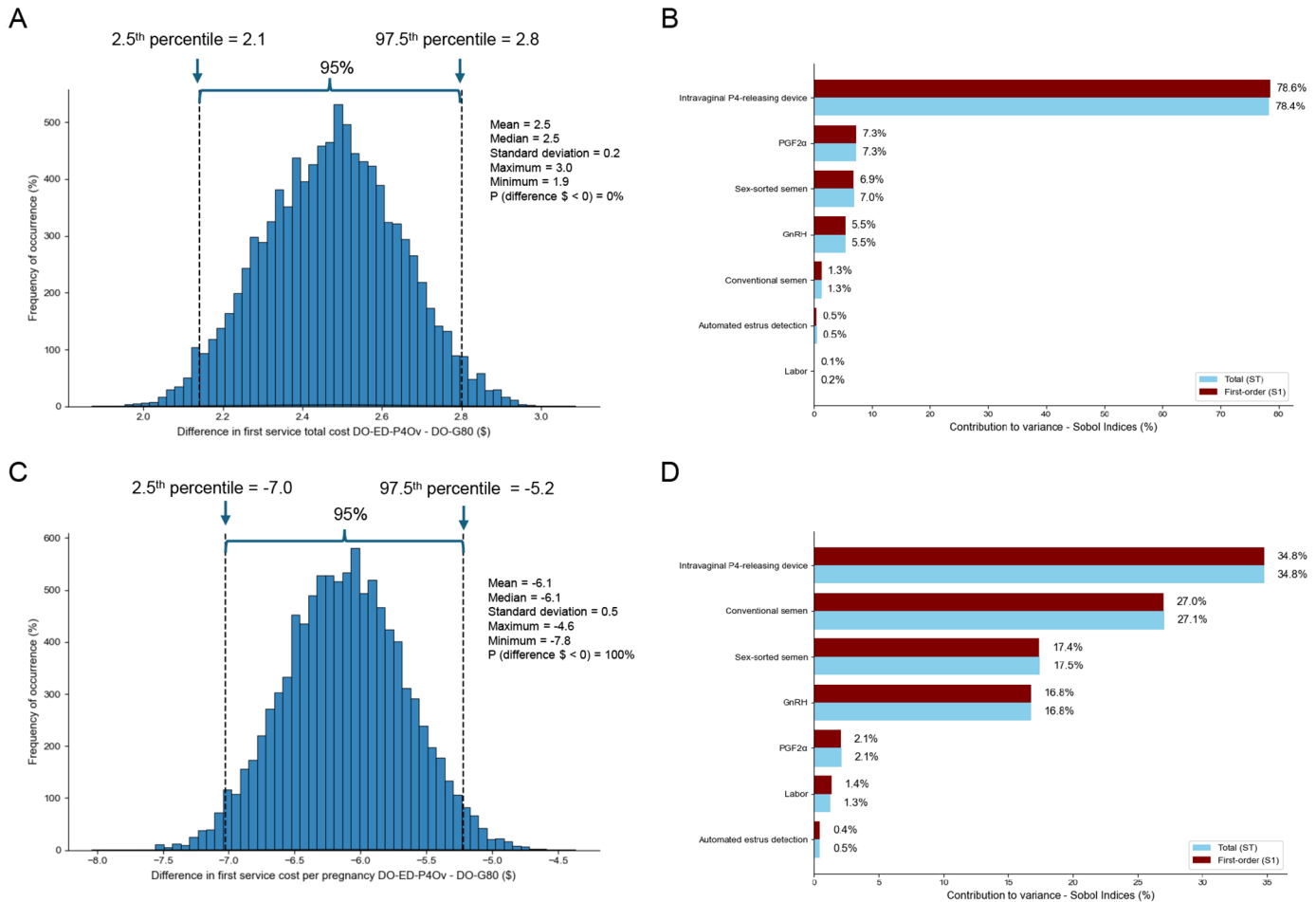


Figure 5. Relative frequency distribution for differences for (A) first-service cost (\$ per cow inseminated) and (C) cost per pregnancy at first AI (\$/pregnancy after first AI) for 10,000 iterations of stochastic simulation. The experimental treatment consisted of submission to first service after synchronization with the Double-Ovsynch protocol with an 80 h interval between induction of luteolysis and induction of ovulation (DO-G80, blue bars) and a targeted reproductive management program including insemination of cows in estrus during a 7-d period after induction of luteolysis in the Double-Ovsynch protocol and submission of nonestrus cows to an Ovsynch protocol with P4 supplementation (DO-ED-P4Ov, green bars). Tornado graphs depict the contribution to the variability in differences between treatments for (B) first-service cost (\$ per cow inseminated) and (D) cost per pregnancy at first AI (\$/cow inseminated and pregnant) of input variables included in the stochastic simulations.

increase was sufficient to increase the overall proportion of cows pregnant after the first service by ~8 percentage points (**p.p.**) and generate more pregnancies by the time all cows received first service despite a wider range of DIM at insemination. Moreover, the P4-Ov protocol for nonestrus cows in the DO-ED-P4Ov program ensured timely insemination (i.e., 10 d after enrollment) and P/AI in the 45% range. Achieving P/AI in this range for nonestrus cows was remarkable considering that the small subgroup of cows (i.e., 12%) that reached P4-Ov was expected to have relatively poor fertility. As most nonestrus cows were not expected to be at an optimal stage of the estrous cycle to start the traditional Ovsynch-56 protocol (i.e., lacked a functional CL and were likely at variable stages of follicle development), P4 supplementation and the 2 PGF_{2 α} treatments aimed

to increase P4 concentrations during ovulatory follicle growth and ensure complete luteolysis and minimal P4 concentrations before TAI (Lima et al., 2009; Pérez et al., 2020). Previously, these interventions were shown effective for enhancing P/AI in lactating cows with no CL at the beginning of Ovsynch (Wijma et al., 2018). Thus, for the DO-ED-P4Ov treatment the combined effect of P/AI in the range of 55% for almost ~90% of the cows that received AIE and 45% P/AI for the ~10% of the cows that received P4-Ov, generated enough pregnancies to fully compensate for the longer and wider range of DIM at first service. Greater P/AI also reduced reproductive costs per pregnancy by ~\$8 despite greater total costs per insemination. As for the effect on P/AI, this positive economic effect was through the decrease in costs per pregnancy for primiparous cows (~\$15) and

a similar cost per pregnancy for multiparous cows. Of note, the results of the stochastic analysis demonstrated a benefit for the DO-ED-P4Ov program under all the market pricing scenarios evaluated. Thus, compared with the DO protocol with delayed induction of ovulation and insemination of cows at detected estrus before TAI, the DO-ED-P4Ov program enhanced fertility and created more pregnancies by the end of the first service period through an effect on the fertility of primiparous cows. A greater pregnancy rate also resulted in reduced reproductive implementation costs.

The GnRH treatment for induction of ovulation before TAI in the DO protocol was removed to increase the proportion of cows that expressed estrus. This enabled the preovulatory follicle of more cows to reach the size needed to secrete enough estradiol to trigger estrous behavior and spontaneous ovulation (Sartori et al., 2001). We hypothesized that expression of estrus would be associated with high P/AI as observed when induction of ovulation was delayed from 56 to 80 h after induction of luteolysis in the DO protocol (Laplacette et al., 2025). In the context of synchronization of ovulation protocols with delayed or no induction of ovulation, expression of estrus is likely an indicator of the response of cows to synchronization, the reproductive potential of the cow at that time, or combinations of both. Indeed, differences for these outcomes manifest through physiological differences during the proestrus period and after ovulation. For example, we previously reported (Laplacette et al., 2025) that nonestrus cows that received TAI in the DO-G80 treatment had smaller follicles before AI, were less likely to ovulate after AI, and had reduced circulating P4 6 d after AI compared with cows that expressed estrus. Fewer nonestrus cows had uterine tone and uterine fluid and had reduced uterine fluid scores compared with estrus cows. These differences were attributed, at least in part, to forcing early ovulation with GnRH. To avoid these issues and promote the self-separation of cows with high versus low reproductive potential (i.e., estrus vs. nonestrus cows), in this experiment cows were provided more time to express estrus after induction of luteolysis. Allowing self-separation of cows with different reproductive potential through expression of estrus also aimed to target the P4-Ov protocol on a small population of cows with low reproductive potential. The use of the P4-Ov protocol also aimed to compensate for the effect other factors with a potential contribution to the poorer fertility of nonestrus cows including multiparity, higher MY, and health events before insemination (Laplacette et al., 2025).

In support of our hypothesis and as expected based on the ovarian and endocrine function of cows during proestrus, removing the GnRH after induction of luteolysis in the DO protocol doubled the proportion of cows that

received AIE and increased the overall proportion of cows that expressed estrus before AI. The distribution of estrus after induction of luteolysis was in line with previous observations (Schilkowsky et al., 2021) as most cows had AEA on d 3 and 4 and a few cows had AEA on d 2, 5, and 6. Our findings also demonstrated that only a modest increase in the proportion of cows that express estrus (~10%) should be expected if GnRH is delayed beyond 80 h of induction of luteolysis or completely removed at the end of the DO protocol. Thus, shorter intervals to initiation of synchronization of ovulation for nonestrus cows could be explored to reduce days to AI without substantially reducing the proportion of cows that receive AIE or express estrus.

As expected, more primiparous than multiparous cows received AIE in the TRM program; however, more cows from both parity groups received AIE (primiparous +42%, multiparous +43%) and had AEA (primiparous +7%, multiparous +12%, data not shown) in the DO-ED-P4Ov than in the DO-G80 treatment. These results demonstrate that TRM strategies aimed at increasing the proportion of cows AIE after the DO protocol can be effective for both primiparous and multiparous cows. In agreement with our previous experiment comparing the DO-G80 to a traditional DO protocol with GnRH 56 h after induction of luteolysis (Laplacette et al., 2025), we observed an interaction between MY group and treatment for the proportion of cows AIE. This interaction can be explained by the association between MY and the dynamics of expression of estrus in each protocol. The decline in the proportion of cows AIE for the high MY group in the DO-G80 treatment was likely due to the positive correlation between time to estrus and greater MY (Laplacette et al., 2025). As greater MY is associated with longer time to estrus, fewer high producing cows had enough time to express estrus before the GnRH treatment. Conversely, the lack of difference for MY groups in the DO-ED-P4Ov treatment was likely due to the 7-d window for estrus detection. Such a long window of opportunity was sufficient for all cows with the potential to exhibit estrus to do so, regardless of the longer interval required by higher milk-yielding groups to display estrous behavior.

Increasing the proportion of cows AIE, increased the overall P/AI for the DO-ED-P4Ov treatment but only due to the differences observed for primiparous cows. Although the number of observations for some days and groups in this experiment was limited, the pattern of P/AI across days and for cows that received AIE versus TAI shed some light on the reasons for differences in P/AI gain for the DO-ED-P4Ov between parity groups. On one end, fewer primiparous (8.8%) than multiparous (13.2%) cows received TAI after the P4-Ov protocol which resulted in 16 and 11 p.p. fewer P/AI than inseminations in

estrus for primiparous and multiparous cows, respectively. This difference was most likely due to a better overall response to the DO protocol and greater estrus expression potential for primiparous than multiparous cows. Differences in fertility and synchronization of ovulation with the DO protocol for primiparous and multiparous cows have been previously reported (Herlihy et al., 2012; Stangaferro et al., 2018). As more cows received TAI and had substantially fewer P/AI, the overall gain in P/AI from AIE services for multiparous cows in the DO-ED-P4Ov treatment was diluted. However, compared with multiparous cows (88%), more primiparous cows (93%) were inseminated on d 3 and 4 after induction of luteolysis when P/AI were numerically greatest and before the substantial decline observed on d 5. More inseminations on d 3 and 4 after induction of luteolysis could have resulted in a more optimal endocrine environment and timing of insemination before ovulation for more primiparous cows. Conversely, multiparous cows in the DO-ED-P4Ov treatment had the longest interval to the onset of estrus which resulted in a larger proportion of cows inseminated after d 4. More time from luteolysis to ovulation resulted in more cows with an extended period of ovulatory follicle dominance, which could have compromised fertility in some cows. Ovulatory follicle exposure to the high LH pulsatility observed after luteolysis in high producing dairy cows with prolonged ovulatory follicle dominance is known to reduce fertility (Cerri et al., 2009; Wijma et al., 2017) through premature oocyte maturation (Revah and Butler, 1996). Future research could explore the potential value of reducing the days of ovulatory follicle dominance to avoid the insemination of cows with oocyte prematuration, especially for multiparous cows. For example, nonestrus cows could be enrolled earlier in the synchronization of ovulation protocol for TAI. However, earlier enrollment in the synchronization protocol for nonestrus cows should be balanced with the potential reduction in cows AIE.

Programs designed to optimize the endocrine environment and ovulatory follicle growth before AI through synchronization of ovulation and expression of estrus are novel and have only been evaluated in a limited number of experiments with lactating dairy cows (DeJarnette et al., 2001; Laplacette et al., 2025). A specific challenge with these programs is that increasing the proportion of cows that express estrus requires delaying induction of ovulation after luteolysis. Ultimately, the subgroup of nonestrus cows before TAI not only does not benefit from expression of estrus but cows in this group also have an extended period of ovulatory follicle dominance. In our previous and this experiment, these cows, which represented ~30% of all cows that received first service, had the lowest P/AI (~25 p.p. less than cows that expressed estrus), which reduced the overall P/AI of the DO-G80

protocol. In the current experiment, the DO-ED-P4Ov treatment was designed to minimize the detrimental effect of nonestrus cows on overall P/AI by minimizing the size of this subgroup and enhancing their fertility. Compared with the DO-G80 protocol in the current and our previous experiment (Laplacette et al., 2025), we confirmed that the proportion of nonestrus cows can be reduced by providing more days for expression of estrus. Moreover, although not directly comparable due to differences in the population of cows that received TAI in the DO-G80 treatment in these experiments, the P4-Ov protocol seemed justified for nonestrus cows after a 7-d period of AIE. Pregnancies per AI in the range of 45% contributed to maintain rather than reduce the overall P/AI of the DO-ED-P4Ov treatment. Conversely, the relatively low P/AI of nonestrus cows that received TAI in the DO-G80 treatment confirmed that these cows typically have poor fertility. Ultimately, the fertility reduction of these cows can reduce the overall P/AI of TRM programs with delayed induction of ovulation to the point of offsetting the value of increasing AIE services.

Evaluation of reproductive costs per AI and costs per pregnancy do not reflect the full complexity of the effects of reproductive programs on dairy herd profitability but offers valuable insights for decision-making. A concern with the DO-ED-P4Ov program is the potential to increase reproductive costs because the P4-Ov protocol is needed for nonestrus cows. This protocol includes 4 additional hormonal treatments and P4 supplementation through an intravaginal P4 releasing device with a cost typically 5- to 10-fold greater than a dose of GnRH or PGF. Part of these extra costs, which only apply to a small proportion of cows, are also offset by reduced GnRH treatment costs as the last GnRH of DO is not used in the DO-ED-P4Ov program. Therefore, the insemination and pregnancy dynamics generated by this program is relevant to the overall cost of implementation. In the current experiment, we observed an increase of \$2.50 per inseminated cow but generating more P/AI resulted in reduced cost per pregnancy by ~\$8 per cow. Thus, as long as the DO-ED-P4Ov program results in a high proportion of cows AIE and greater P/AI than fertility programs that include a relatively similar number of hormonal treatments, reproductive costs are likely to be favorable to the DO-ED-P4Ov program.

From an on-farm implementation perspective, the TRM program combining AIE and TAI with the DO-ED-P4Ov protocol presents some features that might be of value to some farms. For example, the ability to identify cows with different reproductive potential (i.e., high vs. low) through expression of estrus after induction of luteolysis could be used to allocate cows to different insemination strategies. More expensive, and lower fertility sex-sorted semen (DeJarnette et al., 2010; Chebel and

Cunha, 2020; Drake et al., 2020) could be assigned to all or a subpopulation of the high fertility group (i.e., cows that express estrus) whereas less expensive, and greater fertility conventional dairy or beef semen (DeJarnette et al., 2010; Chebel and Cunha, 2020; Drake et al., 2020), could be assigned to the low fertility group. Also, this type of program could be beneficial for dairy farms that prefer to take advantage of effective synchronization of ovulation protocols to optimize P/AI but benefit from or prefer to distribute inseminations over more days of the week compared with the use of TAI. Expanding days to first service by distributing inseminations at detected estrus over 7 d and re-enrolling cows in a synchronization of ovulation protocol could result in a slight delay of first service (as observed in this experiment). However, it is likely that the effect of delaying first insemination is fully offset by the overall increase in fertility, and by the good fertility of the small percentage of cows enrolled in P4-Ov. Identification of a lower reproductive potential group could also offer an opportunity to implement therapies or interventions to increase P/AI in this subgroup of cows. However, fertility-enhancing interventions for nonestrus cows in TRM programs that combine synchronization of ovulation and AIE have not been explored.

Limitations of this Experiment

Notable limitations of this experiment include lack of external validity, a risk of poor reproducibility of some results, and a bias toward greater overall first-service fertility due to implementation of farm-specific management procedures. Lack of external validity stems from conducting the experiment at a single commercial dairy farm where the effect of treatments could have been affected by several environmental and management conditions unique to this farm. In this regard, the results observed might not be repeatable across farms that inseminate cows less frequently than 3×/d after each milking session. Insemination at this high frequency could have resulted in greater P/AI through optimized timing of insemination in most cows (Santos et al., 2025). Strategies such as the DO-ED-P4Ov program should therefore be evaluated in more farms with different management conditions including insemination in estrus 1×/d or 2×/d instead of 3×/d. Alternatively, farms could adapt management to inseminate cows more frequently. The practical and economic implications of the adjustment should be accounted for and quantified to ensure the benefits of improved fertility outweigh the complexity and burden of more frequent inseminations. Identification and treatment of cows with endometritis and a longer voluntary waiting period for primiparous cows could have also influenced first-service outcomes, especially in primiparous cows that received AIE. A longer voluntary waiting period has

been previously associated with improved uterine health, reduced anovulation, and optimized BCS at AI (Stangaferro et al., 2018; Niozas et al., 2019). These conditions individually or collectively are associated with greater first-service P/AI in cows that receive AIE (Niozas et al., 2019). Finally, even though we have previously demonstrated that the DO-G80 protocol resulted in the same overall P/AI than the traditional Double-Ovsynch protocol with 56 h from induction of luteolysis to induction of ovulation, the DO-ED-P4Ov protocol should be compared with fertility protocols (i.e., Double-Ovsynch, G-6-G, or Presynch-Ovsynch) using the typical 56-h interval from induction of luteolysis to induction of ovulation and with no insemination of cows in estrus before TAI. Another limitation of the analyses included in this manuscript is the lack of adjustments for multiple testing across outcomes. Secondary outcomes were included to help explain differences between treatment groups for P/AI, provide a sense of the practical implications of using the experimental treatments on farms, or offer a physiological and logical explanation to findings for the primary outcome. Including multiple analyses could have increased the chances of type I error.

CONCLUSIONS

The DO-ED-P4Ov treatment, which was designed to maximize the proportion of cows AIE after synchronization of ovulation with the DO protocol and enrollment of nonestrus cows in a protocol (i.e., P4-Ov) designed to optimize their fertility, could be an alternative to increase first-service fertility and reduce costs per pregnancy primarily through better performance of primiparous cows. This targeted reproductive program could also have other herd management benefits including identifying cows of different fertility potential and the distribution of AI services over more days of the week compared with submission of all cows to TAI. Compared with the use Double-Ovsynch with induction of ovulation at 80 h after induction of luteolysis, the DO-ED-P4Ov program resulted in insemination of most cows in estrus and increased overall fertility to first service through more P/AI for primiparous cows.

NOTES

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Nonstandard abbreviations used: AEA = automated estrus alert; AI = artificial insemination; AIE = insemination in estrus; CIDR = controlled internal drug release; CL = corpus luteum; DO = Double-Ovsynch; DO-G80 = DO protocol with 80-h interval and AIE between induction of luteolysis after PGF-L and GnRH before TAI; DO-ED-P4Ov = cows were eligible to receive AIE for 7 d after luteolysis with PGF-L of DO and, if not AIE, submitted to P4-Ov; MY = milk yield; OR = odds ratios; P4 = progesterone; P4-Ov = modified Ovsynch protocol with P4 supplementation and 2 PGF_{2α} injections; P/AI = pregnancy per AI; PGF-L = synchronized with a Double-Ovsynch protocol up to the first PGF_{2α}; p.p. = percentage points; TAI = timed artificial insemination; TRM = targeted reproductive management; Trt = treatment.

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